

Unnamed_Unit

Unit 1 Exploring the Core Probability Through Rock-Paper-Scissors

PDF Documents

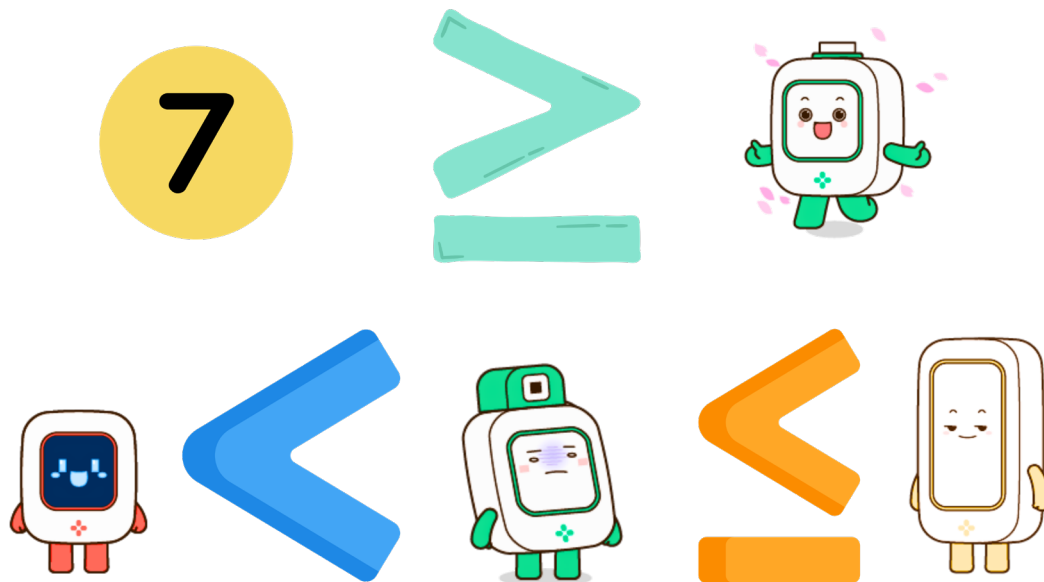
PDF Document 1: 1731845361863-Chapter 7_ unit1 Exploring the Core.pdf

This PDF document is included in the unit materials.



UNIT 1 EXPLORING THE CORE

Topic Investigation



Mathematical ranges are essential for describing quantities that fall within specific limits. These ranges help in everyday decisions and in technological applications. This unit explores how expressions like "greater than or equal to" and "less than or equal to" are used to define these ranges.

UNDERSTANDING RANGES:

A range in mathematics refers to the set of values that lie between two endpoints. These endpoints are numbers that define the beginning and end of the range. We often express ranges using inequalities like $x \geq a$ (greater than or equal to) or $x \leq b$ (less than or equal to).

WHY RANGES MATTER

Understanding and using ranges allows us to create specific guidelines and to ensure that conditions are met, whether we're talking about following rules or making sure a machine operates within safe limits.

TIPS

Did you notice how the math concept of 'range' is similar to using conditions in programming?





NOTATION AND EXAMPLES

Closed Range

When we use the notation $[a,b]$, it means the range includes all numbers from a to b , including a and b themselves. For example, the range of ages that can legally drive could be from 16 to 75 years, written as $[16, 75]$.

$$16 \leq \text{from 16 to 75} \leq 75$$

Closed Range

Notation like (a,b) represents values that are greater than a and less than b but do not include a and b themselves. This might be used when setting temperatures for a chemical reaction that shouldn't reach the freezing or boiling point of the substances involved.

$$16 < \text{from 17 to 74} < 75$$

RANGES IN EVERYDAY LIFE



Cooking Temperatures

When baking, oven temperatures must fall within a specific range, such as 325°F to 375°F, to ensure food cooks properly without burning.

Amusement Park Rides

Rides often have age or height ranges, like needing to be at least 10 years old and 48 inches tall, to ensure safety for all riders.



Inquiry & Reflection

Discussion question 1

Exploration Activity: Math in Action: Think of examples of ranges you see every day, like age limits for movies or temperature settings on appliances. Write down how these ranges help keep things safe or organized.

Discussion question 2

Express your answer as a mathematical expression: A refrigerator must keep food at a safe temperature range between 35°F and 38°F to prevent bacteria growth. Express this temperature range using mathematical inequality symbols.